

Chapter 2 — Variables and Data Types

AI Study Assistant — RAG Knowledge Base

Learning Objectives

Understand variables and assignment.

Identify common Python data types.

Use type conversion and the `type()` function.

Understand strings, numbers, and Boolean values.

1. Variables

A variable is a name that refers to a value. In Python, a variable is created when a value is assigned to it.

```
name = "Ayesha"  
age = 18
```

Python determines the type of the value assigned to a variable; explicit type declarations are usually not required.

2. Naming Variables

Variable names may contain letters, digits, and underscores, but they cannot begin with a digit. Python keywords cannot be used as ordinary variable names.

Good names communicate purpose, such as `student_name` or `total_marks`.

3. Common Data Types

int: whole numbers such as 10 or -4.

float: numbers with a decimal component such as 3.14.

str: text such as "Python".

bool: logical values True or False.

list: an ordered, mutable collection.

tuple: an ordered, immutable collection.

dict: a collection of key-value pairs.

set: an unordered collection of unique values.

4. Strings

Strings represent text and are written inside quotation marks.

```
language = "Python"
```

Strings can be combined using `+` and repeated using `*`.

```
"Py" + "thon" gives "Python".
```

5. Type Conversion

Type conversion changes a value from one type to another when the conversion is valid.

```
int("25")  
float("3.5")  
str(100)
```

Converting user input is common because `input()` returns text.

6. Checking Types

The `type()` function reports the type of an object.

```
x = 10  
print(type(x))
```

The result identifies `x` as an integer.

Key Points for Revision

Assignment uses the `=` operator.

Python variables do not normally require a declared type.

`int`, `float`, `str`, and `bool` are fundamental built-in types.

`input()` returns a string unless the value is explicitly converted.

`type()` can be used to inspect a value's type.